

FLATTRIKE · DESIGN DECISION RECORD

FlatTrike TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	FTK-DDR-001	Version	0.2
Status	Draft	Date	2026-09-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/flattrike		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 decisions on the TRL 2 review and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted (items 1 to 9, 11 and 12); items 13 to 19 accepted later the same day and recorded in FTK-DDR-002; item 10 remains proposed, awaiting Amish

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed eleven items as "Proposed, awaiting Amish", plus a suggested nuance to the problem line. On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." The same instruction approved three cross-cutting SwapCell interface additions, a pricing rule for shared SwapCell packs, and a rule that community designs pick co-design partners per area later.

This record lists what that instruction decides and what it leaves open because there was no recommendation to accept. TRL 4 is on hold by Amish's instruction.

■ Options considered

The options for each item are in docs/REVIEW.md (TRL 2 section) and FTK-PRC-001 v0.2. They are not repeated here.

■ Decision

Table 1. Decided items.

#	Item	Decision	Where it now lives
1	Layout	Decided by Amish, 2026-09-25: go with recommendation. Tadpole front loader (option A)	FTK-PRC-001 v0.3, <code>cad/src/model.py</code>
2	Steering	Decided by Amish, 2026-09-25: go with recommendation. Box steering on one kingpin (option A) for the first prototype, with Ackermann steering (option B) assessed at TRL 3 because of R10. The assessment is in FTK-CAL-001 section 5 and leads to open item 13	FTK-PRC-001 v0.3, FTK-CAL-001
3	Electric assist	Decided by Amish, 2026-09-25: go with recommendation. No assist in the prototype (option A), with the bottom bracket housing able to take a mid-drive later; route C (48 V mid-drive on a SwapCell pack) is the preferred assist route	FTK-PRC-001 v0.3, FTK-REQ-001 R13 and R15, <code>bom/bom.csv</code> line 15
4	Plate material	Decided by Amish, 2026-09-25: go with recommendation. 3 mm S235JR or A36 class steel, primed and painted or powder coated	FTK-PRC-001 v0.3, <code>bom/bom.csv</code>
5	Joint hardware	Decided by Amish, 2026-09-25: go with recommendation. M8 class 8.8 bolts with all-metal prevailing-torque locknuts	FTK-PRC-001 v0.3, <code>bom/bom.csv</code> line 4
6	Head tube and bottom bracket housings	Decided by Amish, 2026-09-25: go with recommendation. Bought steel head tube and bottom bracket shell sections clamped between plates (option A)	FTK-PRC-001 v0.3, FTK-DWG-001
7	Wheels and brakes	Decided by Amish, 2026-09-25: go with recommendation. 20 in (ISO 406) wheels on all three positions with drum brakes; 3-speed drum hub at the rear	FTK-PRC-001 v0.3, FTK-REQ-001 R5
8	R8 and R10 shortfalls	Decided by Amish, 2026-09-25: go with recommendation. Try design changes first at TRL 3 before any target is relaxed. Done: lightening windows, 9 mm box walls, seat moved forward, track widened to 0.84 m. The changes did not close either gap (FTK-CAL-001), so open items 14 and 15 follow	FTK-CAL-001, FTK-REQ-001 R8 and R10
9	Production cost target	Decided by Amish, 2026-09-25: go with recommendation. \$300 or less per trike at 100 units	FTK-REQ-001 new R18
11	Budget	Decided by Amish, 2026-09-25: go with recommendation. Keep <code>budget_usd: 800</code> ; it covers the pedal-only prototype. The assist kit and any SwapCell pack are outside it	<code>project.yaml</code> (unchanged), FTK-REQ-001 R15
12	Problem line	Decided by Amish, 2026-09-25: go with recommendation. Add that cheap local workshop tricycles are heavy, flexible and short-lived, so the case rests on durability, mass, gears and repair as well as price	<code>project.yaml</code> , <code>README.md</code> , FTK-PRB-001 v0.3

Cross-cutting approvals from the same instruction, recorded here:

- **SwapCell interface v0.3.** Decided by Amish, 2026-09-25 (cross-cutting): the SwapCell interface adds a wake method for hosts without CAN (item W), a charge-while-discharging mode (item C) and a latch vibration rating for vehicles (item V, latch class V1). If assist route C is built, FlatTrike cites **SwapCell interface v0.3** items W, C and V: its vehicle receiver meets latch class V1, fits the 10 kΩ coded INTERLOCK loop so a controller without CAN can wake the pack, and leaves the back and lid faces of the pack open to air (SWC-PRC-001 v0.3).

- **Shared packs are priced once.** Decided by Amish, 2026-09-25 (cross-cutting): the SwapCell pack is priced in the SwapCell repo only and excluded from the FlatTrike budget. `bom/bom.csv` line 15 prices the mid-drive kit and receiver only.
- **Co-design partners per area later.** Decided by Amish, 2026-09-25 (cross-cutting): community designs pick co-design partners per area later. Partners stay open.

Items raised at TRL 3

Items 13 to 19 were raised by the TRL 3 calculations (FTK-CAL-001 v0.1). Amish accepted their recommendations on 2026-09-25 ("i accept all your recommendations, go with them across all repos"); FTK-DDR-002 records what changed. Item 10 has no recommendation and stays open.

Table 2. Items raised at TRL 3, and item 10.

#	Item	Status and recommendation
10	First partner organization and city	Proposed, awaiting Amish. No recommendation was made; portfolio rule: partners are chosen per area later
13	Steering for the first prototype after the TRL 3 assessment	Decided by Amish, 2026-09-25: go with recommendation. Switch the first prototype to Ackermann steering (option B) with a fixed box (FTK-DDR-002). Before the decision: Box steering lets the empty trike tip at 5.8 km/h at full lock (0.11 g). Recommendation: switch the first prototype to Ackermann steering (option B, about \$40 more)
14	R8 empty mass (67.5 kg against 55 kg)	Decided by Amish, 2026-09-25: go with recommendation. R8 relaxed to 70 kg for the vehicle with its box (FTK-DDR-002). Before the decision: Options: relax R8 to 70 kg (still lighter than the 80 kg traditional rickshaw), count the box separately from the vehicle, or study aluminium plate (about a third of the plate mass, but fatigue and cost risk). Recommendation: relax R8 to 70 kg for the vehicle with its box
15	R10 with no cargo (0.24 g against 0.30 g)	Decided by Amish, 2026-09-25: go with recommendation. R10 rider-only case relaxed to 0.22 g together with Ackermann steering and a cornering-speed label (FTK-DDR-002). Before the decision: Options: relax the rider-only case to 0.22 g with a cornering-speed label, specify 30 kg of low ballast when running empty, or accept a 1.0 m track and break R9. Recommendation: relax the rider-only case to 0.22 g together with Ackermann steering (item 13)
16	R1 with a 90 kg rider (307.5 kg gross)	Decided by Amish, 2026-09-25: go with recommendation. Cargo rated at 140 kg for riders over 80 kg, keeping the 300 kg class limit (FTK-DDR-002). Before the decision: Recommendation: rate the cargo at 140 kg for riders over 80 kg, keeping the 300 kg class limit
17	R14 spine side plate repair (about 78 min)	Decided by Amish, 2026-09-25: go with recommendation. R14 accepts 90 min for the spine side plates only (FTK-DDR-002). Before the decision: Recommendation: accept 90 min for the spine side plates only, since they are the least likely plates to be damaged
18	R18 production cost (about \$414 against \$300)	Decided by Amish, 2026-09-25: go with recommendation. Keep the \$300 target and get wholesale prices for wheels and hubs from the first partner before any change (FTK-DDR-002; waits on item 10). Before the decision: Recommendation: keep the target and get wholesale prices for wheels and hubs from the first partner before any change
19	Headset under roll load (3.34 kN per bearing)	Decided by Amish, 2026-09-25: go with recommendation. Longer head tube, 150 to 200 mm; taper roller bearings not needed once the bed is fixed (FTK-DDR-002). Before the decision: Recommendation: a longer head tube or taper roller bearings, sized in the FEA step

■ Consequences

- FTK-PRB-001, FTK-PRC-001 and FTK-REQ-001 move to v0.3 with these decisions; the precis no longer lists items 1 to 9 as proposed.
- FTK-REQ-001 R15 now states that the \$800 budget covers the pedal-only prototype, and a new R18 holds the production cost target.
- The TRL 3 model, drawing FTK-DWG-001 and FTK-CAL-001 follow the decided layout, steering, materials and hardware.
- Items 13 to 19 are implemented at TRL 3 as set out in FTK-DDR-002.
- TRL 4 work (building, testing, purchasing) is on hold by Amish's instruction, whatever the outcome of the open items.